Migration Framework Overview

> **Series:** M2S | **Notebook:** 2 of 8 | **Created:** January 2026 | **Last Updated:** 03/02/2026

Migrating from Dynatrace Managed to SaaS is not a single action—it's a journey. The Dynatrace Migration Framework provides a structured approach that has been validated across hundreds of successful migrations.

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Prerequisites

Before starting this notebook, you should have:

Requirement	Description
SaaS tenant provisioned	Access to your new Dynatrace SaaS environment
Environment assessment	Basic inventory of your Managed deployment
Stakeholder alignment	Team readiness for migration

Learning Objectives

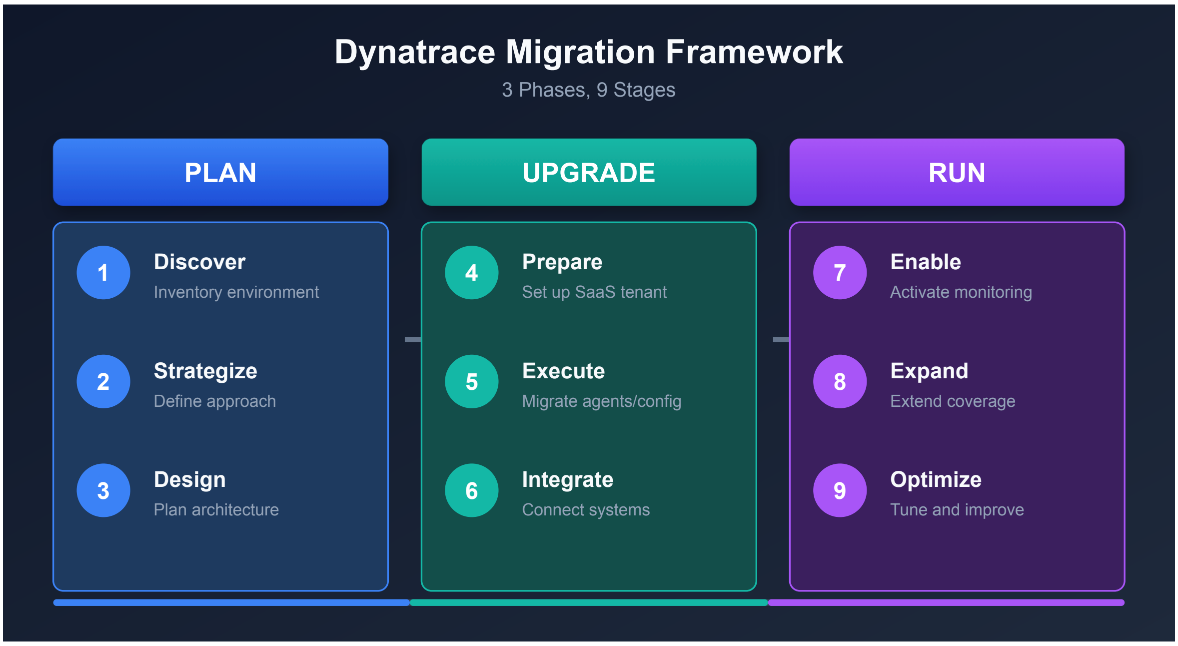
By the end of this notebook, you will:

- Understand the three-phase migration framework
- Know the stages within each phase
- Be able to identify key activities for each stage
- Choose between Big Bang and Phased migration approaches

1. Introduction

Why a Framework Matters

Without Framework	With Framework
Ad-hoc decisions	Structured approach
Missed configurations	Complete migration
Extended downtime	Minimized disruption
Unclear ownership	Defined responsibilities
Reactive troubleshooting	Proactive validation



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2. The Three-Phase Framework

The migration framework consists of three phases, each with three stages:

Framework Overview

Phase	Stage 1	Stage 2	Stage 3
Plan	Discover	Strategize	Design
Upgrade	Prepare	Execute	Integrate
Run	Enable	Expand	Optimize

Time Investment by Phase

Phase	Typical Duration	Effort Level
Plan	2-4 weeks	Medium
Upgrade	1-4 weeks	High
Run	2-4 weeks	Medium

> **Note:** Duration varies significantly based on environment size and complexity.

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3. Phase 1: Plan

The Plan phase ensures you understand what you have and how you'll migrate it.

Stage 1: Discover

Objective: Create a complete inventory of your current environment.

> **Critical Planning Constraint:** SaaS tenants have a maximum of 25,000 monitored hosts. If consolidating multiple Managed clusters, plan for multiple SaaS tenants if the combined host count exceeds this limit.

Activity	Deliverable
Enumerate monitored entities	Host, service, application counts
Document configurations	Settings export
Identify integrations	Webhook and API list
Map dependencies	Integration diagram
Review custom code	Extensions, plugins inventory

Discovery Queries

```
// Get complete entity counts by type
fetch dt.entity.host | summarize hosts = count()
| append [fetch dt.entity.service | summarize services = count()]
| append [fetch dt.entity.application | summarize applications = count()]
| append [fetch dt.entity.synthetic_test | summarize synthetics = count()]
```

Stage 2: Strategize

Objective: Define your migration approach and success criteria.

Decision	Options	Considerations
Migration approach	Big Bang vs. Phased	Environment size, risk tolerance
Timeline	Weeks to months	Resource availability
Rollback plan	Full vs. partial	Business criticality
Success criteria	Metrics to validate	Coverage, performance

Key Questions to Answer

1. **What is your risk tolerance?** (Determines approach)
2. **What is your maintenance window?** (Determines timeline)
3. **Who are the stakeholders?** (Determines communication plan)
4. **What are the success criteria?** (Determines validation approach)

Stage 3: Design

Objective: Create the technical migration plan.

Design Area	Considerations
Network architecture	ActiveGate placement, firewall rules
Security model	Token scopes, user permissions
Configuration mapping	What migrates, what's recreated
Testing strategy	Validation checkpoints

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4. Phase 2: Upgrade

The Upgrade phase is where the actual migration happens. The **SaaS Upgrade Assistant** is the primary tool for this phase.

Stage 4: Prepare

Objective: Set up the SaaS environment and prepare for migration.

Activity	Details
Provision SaaS tenant	Work with Dynatrace team
Configure network	Firewall rules, proxy settings
Create API tokens	With required scopes
Set up users	IAM configuration
Prepare ActiveGates	For agent routing
Install SaaS Upgrade Assistant	**On target SaaS tenant, assign `upgrade-assistant:environments:write` IAM policy**

Network Preparation Checklist

Endpoint	Purpose	Port
`{tenant}.live.dynatrace.com`	SaaS cluster	443
`{tenant}.apps.dynatrace.com`	Apps and platform services	443

Stage 5: Execute

Objective: Perform the actual migration.

The SaaS Upgrade Assistant automates the majority of configuration migration:

Activity	Sequence	Tool
Deploy Environment ActiveGate	First	Manual
Export configuration from Managed	**Second**	**SaaS Upgrade Assistant**
Upload and review in SaaS	**Third**	**SaaS Upgrade Assistant**
Deploy configurations (selective import)	**Fourth**	**SaaS Upgrade Assistant**
Redirect OneAgents	Fifth	Manual / automated
Verify data flow	Sixth	DQL validation
Migrate synthetic monitors	Seventh	Manual

> **Tip:** The SaaS Upgrade Assistant supports **selective import**—you can deploy configurations in waves, choosing which configuration types to include or exclude per deployment. Smart dependency management automatically adds or removes dependent configurations.

Stage 6: Integrate

Objective: Connect external systems and validate integrations.

Integration Type	Migration Action
ITSM webhooks	Update URLs to SaaS endpoints
CI/CD pipelines	Update API endpoints and tokens
Custom dashboards	SaaS Upgrade Assistant or recreate
Third-party tools	Reconfigure connections

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5. Phase 3: Run

The Run phase ensures long-term success after migration.

Stage 7: Enable

Objective: Activate full monitoring capabilities.

Activity	Purpose
Enable all monitoring	Full coverage
Configure alerting	Problem notifications
Set up dashboards	Operational visibility
Train users	SaaS UI familiarity

Stage 8: Expand

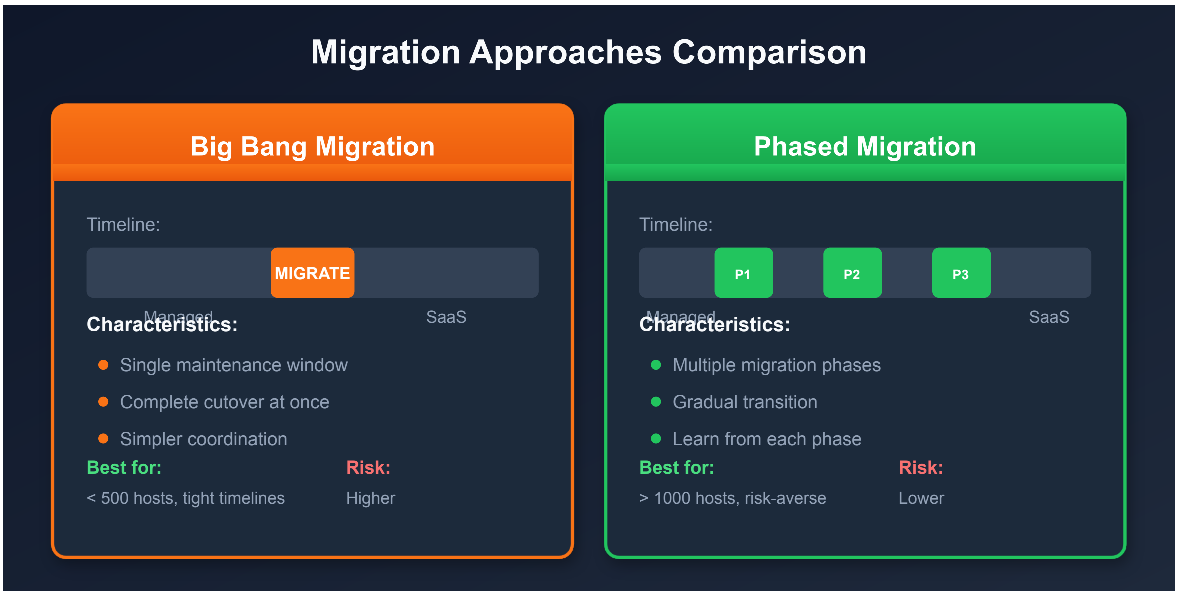
Objective: Extend monitoring to additional areas.

Expansion Area	Consideration
Additional environments	Dev, staging, DR
New applications	Recently deployed
Cloud integrations	AWS, Azure, GCP
OpenTelemetry	Custom instrumentation

Stage 9: Optimize

Objective: Tune for maximum value.

Optimization	Approach
Alert tuning	Reduce noise, increase signal
Query performance	Optimize DQL queries
Cost management	Right-size retention
Feature adoption	Leverage new SaaS capabilities



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6. Migration Approaches

Big Bang Migration

All agents and configurations migrate in a single maintenance window.

Pros	Cons
Faster completion	Higher risk
Simpler coordination	Larger impact if issues
Clean cutover	Requires longer window
No parallel operation cost	All-or-nothing

Best for:
- Smaller environments (< 500 hosts)
- Organizations with flexible maintenance windows
- Teams comfortable with higher risk

Phased Migration

Migration happens in stages, often by environment, region, or application.

Pros	Cons
Lower risk	Longer duration
Lessons learned apply to later phases	Parallel operation complexity
Smaller impact if issues	More coordination needed
Flexible timing	Temporary split visibility

Best for:
- Larger environments (> 1,000 hosts)
- Risk-averse organizations
- Complex integrations
- Limited maintenance windows

Phased Migration Patterns

Pattern	Description	Example
By environment	Dev → Staging → Prod	Lowest risk
By region	EMEA → APAC → Americas	Geographic isolation
By application	Non-critical → Critical	Business priority
By technology	Linux → Windows	Technical grouping

Decision Matrix

Factor	Big Bang	Phased
Hosts < 500	✅ Recommended	⚠️ Optional
Hosts > 1,000	⚠️ Possible	✅ Recommended
Simple integrations	✅ Recommended	⚠️ Optional
Complex integrations	⚠️ Risky	✅ Recommended
Short timeline	✅ Faster	⚠️ Longer
Risk-averse	⚠️ Higher risk	✅ Lower risk

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6 Best Practices for Migration

Based on hundreds of successful migrations, these practices ensure success:

1. Engage Early

Involve Dynatrace Professional Services and your account team early in planning.

2. Assess Thoroughly

Complete discovery before making migration decisions. Unknown dependencies cause issues.

3. Plan Configuration

Document every setting that needs migration. Use the SaaS Upgrade Assistant to export and review all configurations before deploying.

4. Test Incrementally

Validate each phase before proceeding. The SaaS Upgrade Assistant's **selective import** supports a waved approach—deploy configuration types incrementally.

5. Communicate Broadly

Keep all stakeholders informed. Surprises damage trust.

6. Validate Completely

Verify all data flows post-migration. Use the SaaS Upgrade Assistant's deployment result downloads (CSV) alongside DQL validation queries to confirm expected data.

Lessons Learned from Customer Migrations

Real-world migrations have revealed important insights:

The 90/10 Rule

> **Key Learning:** 90% of configurations can be migrated automatically using the SaaS Upgrade Assistant and Terraform/Monaco. The remaining 10% (manual configurations, credentials, and custom scripts) often takes 90% of the time.

The SaaS Upgrade Assistant handles the bulk of this—dashboards, settings, alert policies, and more. Focus your manual effort on non-portable items like Credentials Vault entries, problem notification webhooks, and cloud platform integration credentials.

Tenant Consolidation Limits

When consolidating multiple Managed tenants into a single SaaS tenant:
- **Maximum host count limit: 25,000** per tenant
- If this limit is exceeded, split into multiple SaaS tenants
- The SaaS Upgrade Assistant may encounter issues with **identical configuration names** when consolidating tenants—use Monaco instead in this scenario

Version Alignment

> **Important:** Align your Managed cluster and SaaS environment to the **same major version** (e.g., both 1.294.x) when using the SaaS Upgrade Assistant. Version mismatches cause false-positive configuration failures.

SSO/SAML Configuration

> 🚨 **Critical:** Your Identity Provider (IdP) must sign the **entire SAML message**, not just the assertion. Failure causes authentication errors. Azure AD meets Dynatrace's SAML requirements.

IdP Consideration	Requirement
SAML message signing	Full message, not just assertion
Group limit (Azure Entra)	150 groups per user in SAML claim
Group filtering	Filter to Dynatrace-related groups only

OneAgent Version Requirements

- Support for OneAgent versions is **9-12 months**
- Review oldest supported versions before migration
- Newer versions provide more configuration abilities and simpler re-homing

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7. Next Steps

Immediate Actions

1. **Choose your approach** - Big Bang or Phased based on your environment
2. **Identify phases** - If phased, define the migration waves
3. **Build timeline** - Rough schedule for each phase
4. **Assign ownership** - Who leads each stage?
5. **Install the SaaS Upgrade Assistant** - Prepare on your target SaaS tenant

Continue the Series

Next Notebook	Focus
M2S-03: Planning & Assessment	Deep dive into discovery and planning

Additional Resources

- SaaS Upgrade Assistant Documentation
- Upgrading from Dynatrace Managed to SaaS
- Dynatrace Documentation

- Settings API
- ActiveGate Deployment

Summary

In this notebook, you learned:

- The three-phase migration framework (Plan, Upgrade, Run)
- The nine stages within the framework
- How the SaaS Upgrade Assistant fits into Phase 2 (Upgrade)
- Key activities for each stage
- How to choose between Big Bang and Phased approaches
- The 6 best practices for successful migration

> **Key Takeaway:** A structured framework reduces risk and increases success probability. The SaaS Upgrade Assistant is your primary tool during the Upgrade phase—it automates configuration export, import, and tracking.

Continue to M2S-03: Planning & Assessment for detailed guidance on the Discover, Strategize, and Design stages.
